

1.1.c. Movement analysis

Learners will develop their knowledge of the three classes of lever and will be able to use examples from physical activities and sport to show where these levers might operate to produce movement. Learners will become aware of the mechanical advantage provided by levers in movement.

Learners will know the three planes of movement and be able to give examples of these levers from different physical activities and sports. Frontal, transverse and longitudinal axes of rotation will be recognised by learners who will be able to apply these to examples from physical activities and sports.

Topic area	Learners must:
Lever systems 	• know the three classes of lever and their use in physical activity and sport: ○ 1st class – neck ○ 2nd class – ankle ○ 3rd class – elbow • know the definition of mechanical advantage.
Planes of movement and axes of rotation 	• know the location of the planes of movement in the body and their application to physical activity and sport: ○ frontal ○ transverse ○ sagittal • know the location of the axes of rotation in the body and their application to physical activity and sport: ○ frontal ○ transverse ○ longitudinal.